an upper hand in the region's water geography," says Kugelman.

India is also wary of the timing of the Medog dam approval, which came just as New Delhi and Beijing resumed talks after a prolonged diplomatic freeze post-Doklam (2017) and Galwan (2020). Many analysts see it as a calculated move to pressure India in border negotiations. By intertwining territorial disputes with transboundary rivers, Beijing is likely angling for strategic concessions.

A Himalayan Blunder?

As debate rages over the potential impact of the Medog Dam on India and Bangladesh, one fact is undisputed: its construction poses irreversible ecological risks to the fragile Himalayan region. Dams can trigger tremors, a phenomenon known as reservoir-induced seismicity—when the immense weight of impounded water destabilises fault lines of the Earth. "The sheer weight of a vast reservoir could induce small earthquakes by exerting pressure on the Earth's crust. In an already hyper-seismic zone, the consequences of such stress remain dangerously unpredictable," warn Robert Wasson, adjunct professor, College of Science and Engineering, James Cook University, Australia, and emeritus professor, Fenner School of Environment and Society Australian National University, and Shukla Acharjee, assistant professor, Centre for Studies in Geography, Dibrugarh University.

Beyond seismic threats, the dam's sheer scale demands mass deforestation, stripping the region of its green cover and natural stabilisers. Without tree roots anchoring the soil, heavy monsoon rains could trigger deadly mudslides, raising the risk of dam failure.

The Himalayas, already crumbling under climate stress, have seen a surge in Glacial Lake Outburst Floods, avalanches and landslides. "On March 22, 2021, a massive glacier collapsed in the Sedongpu River basin, on the left bank of the Grand Canyon of the Yarlung Tsangpo and blocked the river and caused water levels to rise by 10 metres," says Modak. Wasson and Acharjee note that the Tsangpo Gorge is one of the most geologically dynamic—and possibly the most active—regions on Earth. "It experiences extreme floods, with peak flows reaching up to a million cubic metres per second. A single such flood can erode as much of the gorge as 4,000 years' worth of annual flow," they add.

The region's volatile geo-morphology also makes dam stability a growing concern. Earthquakes, heavy siltation and landslides will shorten the dams' lifespan. After the January 7 earthquake, inspections of 14 hydropower dams in Tibet found structural cracks in five, forcing three to be emptied. "This region is the most sediment-rich and sediment-producing area on Earth. As a result, dams here degrade faster, increasing the risk for downstream populations," says Ruth Gamble, deputy director (research), La Trobe Asia, La Trobe University, Australia, who specialises in the environmental, cultural and climate history of Tibet, the Himalayas and Asia.

On its part, China insists the Medog project is a clean energy game-changer, not a geopolitical weapon. Wang Lei, chargé d'affaires at the Chinese Embassy in India, dismisses concerns about adverse

HOW INDIA SHOULD RESPOND

THE MANY WAYS IN WHICH INDIA CAN COUNTER CHINA'S WATER DOMINANCE


ALAMY

- Instead of relying on temporary MoUs with China, India must push for a legally binding treaty that guarantees fair and transparent water distribution
- An alliance with lower riparian nations like Bangladesh, Bhutan and Nepal can increase diplomatic pressure on China to adopt more responsible water-sharing policies. India should demand year-round hydrological data-sharing rather than just seasonal flood reports to prevent unexpected surges or shortages
- India needs to strengthen its river management systems by investing in advanced flood forecasting, efficient drainage networks and climate-resilient infrastructure to safeguard vulnerable regions. Cutting-edge surveillance technology should be deployed to monitor Chinese dam activities in real-time and identify any unusual changes in water flow
- To counter China's growing influence over the Brahmaputra, India must expedite the construction of the Siang



"We have consistently raised concerns over China's mega river projects. India will continue pressing China to take necessary measures to protect our interests"

RANDHIR JAISWAL *Spokesperson, Ministry of External Affairs*